

A Candidate Full Solution to the Finite- p Hausdorff Spaceability Question

Packaged for the FA/Banach discovery run

13 June 2026

Source and Target

Source paper: Jia Liu, Saisai Shi, and Zhenliang Zhang, *On strong spaceability of continuous functions and fractal dimensions*, arXiv:2605.25037v2.

The relevant open problem is Question 3.19 on page 24 of the source paper. The packet includes the source PDF as `source_paper.pdf`; the following crop is rendered from that PDF.

can be easily reformulated for the space $C(K)$, where K is a compact subset of $[0, 1]$ with positive Hausdorff measure and $s \in (1, \dim_H K + 1]$.

The following two problems remain open.

Question 3.19. *Is the set $H_s[0, 1]$ $(p, \mathfrak{c})$ -spaceable for integers $3 \leq p < \aleph_0$?*

Question 3.20. *Is the set $\underline{B}_s[0, 1]$ (α, β) -lineable/spaceable for any two cardinal numbers $1 < \alpha < \beta$?*

4. $(\alpha, \mathfrak{c})$ -SPACEABILITY OF FUNCTIONS FOR UPPER BOX DIMENSION

For $1 < s \leq 2$, put

$$H_s[0, 1] = \{f \in C[0, 1] : \dim_H G_f([0, 1]) = s\}.$$

Question 3.19 asks whether $H_s[0, 1]$ is $(p, \mathfrak{c})$ -spaceable for every integer $3 \leq p < \aleph_0$.

Definition. *Let X be a topological vector space, $A \subset X$, and $p \in \mathbb{N}$. The set A is $(p, \mathfrak{c})$ -spaceable if, whenever E is a p -dimensional linear subspace of X with*

$$E \subset A \cup \{0\},$$

there is a closed linear subspace $Y \subset X$ such that

$$E \subset Y \subset A \cup \{0\} \quad \text{and} \quad \dim Y = \mathfrak{c}.$$

Status of This Packet

This is a candidate full solution to Question 3.19. The proof was supplied as an external draft and then repackaged here in the run's standard format. The mathematical argument below makes one correction to the draft's source pointer: the finite-dimensional compact-selection input is Proposition 3.10 of the source paper, taken with $m = 1$, not Proposition 3.4.

Bounded novelty check performed for packaging used the phrases `Question 3.19`, `2605.25037`, `Hs[0,1]`, `(p,c)-spaceable`, and the source-paper title. The search found the source arXiv paper

and no separate later answer. Because this source is very recent, the novelty confidence is moderate rather than high; human review should still check for closely related preprints using the same compact-set interpolation idea.

Imported Result

For a set $K \subset [0, 1]$, write $u \in H_s(K)$ to mean $\dim_H G_u(K) = s$.

Imported Proposition (Liu–Shi–Zhang, Proposition 3.10 with interval domain and $m = 1$). *Let $1 < s \leq 2$, let $p \in \mathbb{N}$, and let $f_1, \dots, f_p \in H_s[0, 1]$ be linearly independent. Assume*

$$\text{span}\{f_1, \dots, f_p\} \subset H_s[0, 1] \cup \{0\}.$$

Then there is a compact set $K_0 \subset [0, 1]$ such that

$$\dim_H K_0 = s - 1$$

and, for every nonzero $(a_1, \dots, a_p) \in \mathbb{R}^p$,

$$\left. \sum_{i=1}^p a_i f_i \right|_{[0,1] \setminus K_0} \in H_s([0, 1] \setminus K_0).$$

Auxiliary Facts

We use the following standard facts.

(F1) Hausdorff dimension is countably stable:

$$\dim_H \left(\bigcup_{n=1}^{\infty} A_n \right) = \sup_n \dim_H A_n.$$

(F2) If $K \subset \mathbb{R}$ and $f \in C(K)$, then

$$\dim_H G_f(K) \leq \dim_H K + 1.$$

Indeed, $G_f(K) \subset K \times f(K)$ and $f(K) \subset \mathbb{R}$.

(F3) If $K \subset [0, 1]$ is compact and $d = \dim_H K$, then K has a full-dimensional point: there is $x_0 \in K$ such that

$$\dim_H(K \cap (x_0 - r, x_0 + r)) = d, \quad r > 0.$$

Otherwise finitely many relatively open neighborhoods of dimension $< d$ would cover K , contradicting finite stability.

(F4) If $E \subset \mathbb{R}$ is an uncountable compact set, then there exists $u \in C(E)$ such that

$$\dim_H G_u(E) = \dim_H E + 1.$$

This is the prevalent maximal graph-dimension theorem cited as Lemma 2.8 in the source paper.

A Compact c_0 -Subspace Lemma

Lemma. *Let $K \subset [0, 1]$ be compact and let $d = \dim_H K > 0$. There is a closed subspace $X_K \subset C[0, 1]$, isomorphic to c_0 , with the following properties.*

(i) *For every nonzero $h \in X_K$,*

$$\dim_H G_h(K) = d + 1.$$

(ii) *Every $h \in X_K$ is affine on each connected component of $[0, 1] \setminus K$, with constant extension on endpoint components if they occur.*

Proof. Choose a full-dimensional point $x_0 \in K$. Let $r_m \downarrow 0$, and put

$$A_m = K \cap ((x_0 - r_m, x_0 + r_m) \setminus [x_0 - r_{m+1}, x_0 + r_{m+1}]).$$

The annuli are pairwise disjoint. Since

$$K \cap (x_0 - r_N, x_0 + r_N) \subset \{x_0\} \cup \bigcup_{m \geq N} A_m$$

has Hausdorff dimension d , countable stability gives

$$\sup_{m \geq N} \dim_H A_m = d, \quad N \in \mathbb{N}.$$

Passing to a subsequence and using inner regularity of Hausdorff dimension, choose compact sets $E_n \subset K$ and open intervals I_n such that

$$E_n \subset I_n, \quad \overline{I_n} \cap \overline{I_m} = \emptyset \ (n \neq m), \quad I_n \rightarrow x_0,$$

and

$$d_n := \dim_H E_n > d - \frac{1}{n}.$$

Discarding finitely many terms, assume $d_n > 0$ for all n . Then $d_n \rightarrow d$, and each E_n is uncountable.

By (F4), choose $\varphi_n \in C(E_n)$ with

$$\dim_H G_{\varphi_n}(E_n) = d_n + 1, \quad \|\varphi_n\|_\infty = 1.$$

Tietze's extension theorem gives $\psi_n \in C(K)$ such that

$$\psi_n|_{E_n} = \varphi_n, \quad \psi_n|_{K \setminus I_n} = 0, \quad \|\psi_n\|_\infty = 1.$$

Partition $\mathbb{N}$ into pairwise disjoint infinite sets $\mathbb{N} = B_1 \cup B_2 \cup \dots$. For $n \in B_j$, choose coefficients $c_{j,n} \in (0, 1]$ such that

$$\sup_{n \in B_j} c_{j,n} = 1, \quad c_{j,n} \rightarrow 0 \ (n \rightarrow \infty, n \in B_j).$$

Define $v_j \in C(K)$ by

$$v_j(x) = \begin{cases} c_{j,n} \psi_n(x), & x \in K \cap I_n \text{ for some } n \in B_j, \\ 0, & \text{otherwise.} \end{cases}$$

The only possible accumulation point of the intervals I_n is x_0 , and the coefficients tend to 0 along each B_j , so v_j is continuous on K and $\|v_j\|_\infty = 1$.

Define

$$T : c_0 \longrightarrow C(K), \quad T((a_j)_{j \geq 1}) = \sum_{j=1}^{\infty} a_j v_j.$$

The sum is pointwise locally finite away from x_0 . Continuity at x_0 follows from $a_j \rightarrow 0$ and $c_{j,n} \rightarrow 0$ on each fixed block. Because the supports of the v_j 's are disjoint,

$$\|T(a)\|_{\infty} = \sup_j |a_j|.$$

Thus T is an isometric embedding and $T(c_0)$ is closed in $C(K)$.

If $a = (a_j) \neq 0$, choose j_0 with $a_{j_0} \neq 0$. For every $n \in B_{j_0}$, on E_n we have

$$T(a) = a_{j_0} c_{j_0,n} \varphi_n.$$

Nonzero scalar multiplication does not change Hausdorff dimension, hence

$$\dim_H G_{T(a)}(E_n) = d_n + 1, \quad n \in B_{j_0}.$$

Since every infinite block B_{j_0} contains indices tending to infinity,

$$\dim_H G_{T(a)}(K) \geq \sup_{n \in B_{j_0}} (d_n + 1) = d + 1.$$

The reverse inequality is (F2), so $\dim_H G_{T(a)}(K) = d + 1$.

Finally, extend functions linearly over every bounded component of $[0, 1] \setminus K$ and constantly over endpoint components. This canonical operator $\mathcal{E}_K : C(K) \rightarrow C[0, 1]$ is a linear isometric embedding. Put

$$X_K = \mathcal{E}_K(T(c_0)).$$

Then X_K is closed in $C[0, 1]$, isomorphic to c_0 , satisfies the same graph-dimension conclusion on K , and is affine on the complementary components. $\square$

Main Theorem

Theorem. *Let $1 < s \leq 2$. Then $H_s[0, 1]$ is $(p, \mathfrak{c})$ -spaceable for every finite $p \in \mathbb{N}$. In particular, Question 3.19 has an affirmative answer.*

Proof. Let $E \subset C[0, 1]$ be a finite-dimensional subspace such that

$$E \subset H_s[0, 1] \cup \{0\}.$$

Let $p = \dim E$ and choose a basis $f_1, \dots, f_p$ of E . Applying the imported proposition gives a compact set $K_0 \subset [0, 1]$ such that

$$\dim_H K_0 = s - 1$$

and, for every $0 \neq w \in E$,

$$w|_{[0,1] \setminus K_0} \in H_s([0, 1] \setminus K_0).$$

Set $d = s - 1 > 0$. Apply Lemma to K_0 , and write

$$X_0 = X_{K_0}.$$

Thus X_0 is closed, $\dim X_0 = \mathfrak{c}$, every nonzero $h \in X_0$ has

$$\dim_H G_h(K_0) = s,$$

and every $h \in X_0$ is affine on the connected components of $[0, 1] \setminus K_0$.

First, $E \cap X_0 = \{0\}$. Indeed, if $0 \neq w \in E \cap X_0$, then the choice of K_0 gives

$$\dim_H G_w([0, 1] \setminus K_0) = s.$$

But $w \in X_0$ is affine on each connected component of the open set $[0, 1] \setminus K_0$. This open set is a countable union of intervals, and the graph of an affine function on each interval has Hausdorff dimension 1. Therefore

$$\dim_H G_w([0, 1] \setminus K_0) \leq 1,$$

contradicting $s > 1$.

Let

$$Y = E \oplus X_0.$$

Since E is finite-dimensional and X_0 is closed, Y is a closed subspace of $C[0, 1]$. Moreover $E \subset Y$ and $\dim Y = \mathfrak{c}$.

It remains to prove $Y \subset H_s[0, 1] \cup \{0\}$. Let $0 \neq y \in Y$ and write uniquely

$$y = w + h, \quad w \in E, \quad h \in X_0.$$

Assume first that $w \neq 0$, and put $U = [0, 1] \setminus K_0$. On every connected component J of U , the function h is affine, so the shear

$$(x, t) \mapsto (x, t + h(x))$$

is bi-Lipschitz on $J \times \mathbb{R}$. Hence

$$\dim_H G_{w+h}(J) = \dim_H G_w(J).$$

Using countable stability over the components of U ,

$$\dim_H G_y(U) = \dim_H G_w(U) = s.$$

On K_0 , (F2) gives

$$\dim_H G_y(K_0) \leq \dim_H K_0 + 1 = s.$$

Thus

$$\dim_H G_y([0, 1]) = \max\{\dim_H G_y(K_0), \dim_H G_y(U)\} = s.$$

Assume now that $w = 0$. Then $y = h \neq 0$, and Lemma gives

$$\dim_H G_y(K_0) = s.$$

The graph over $[0, 1] \setminus K_0$ is a countable union of affine graphs, so it has dimension at most 1. Since $s > 1$, the whole graph has dimension exactly s .

Therefore every nonzero element of Y belongs to $H_s[0, 1]$. The initial finite-dimensional subspace E was arbitrary, so $H_s[0, 1]$ is $(p, \mathfrak{c})$ -spaceable for every finite p . $\square$

Verification Notes

- No numerical verification is used; the proof is purely analytic.
- The crop was generated by

```
conda run --no-capture-output -n sandbox python code/make_open_problem_crop.py.
```

- The main human-review pressure points are: the c_0 -subspace lemma on an arbitrary compact set K with $\dim_H K > 0$; the correction from the draft's Proposition 3.4 pointer to Proposition 3.10 with $m = 1$; and the use of affine interpolation to make the added c_0 -space invisible on the complement of K_0 .
- The original colleague-supplied files are retained in this folder under the names `provided_solution_draft.tex` and `provided_solution_draft.pdf`.